

Python HW

There no time limit for this HW

- In a markdown cell write down the AI model you use if needed
- This time work until you get stuck. Then ask AI for help. If you asked AI for help write down what you asked for help on.
- Clearly state when and where you ask for help.
- If you get a question to work no need to ask AI for help.
- You should be able to use your class notes and notebooks to do all of this work. Try and use your notes. Do not ask AI for help unless really needed.
- Do NOT ask AI to do the question for you. It will show. Use what you know.

AI Model:

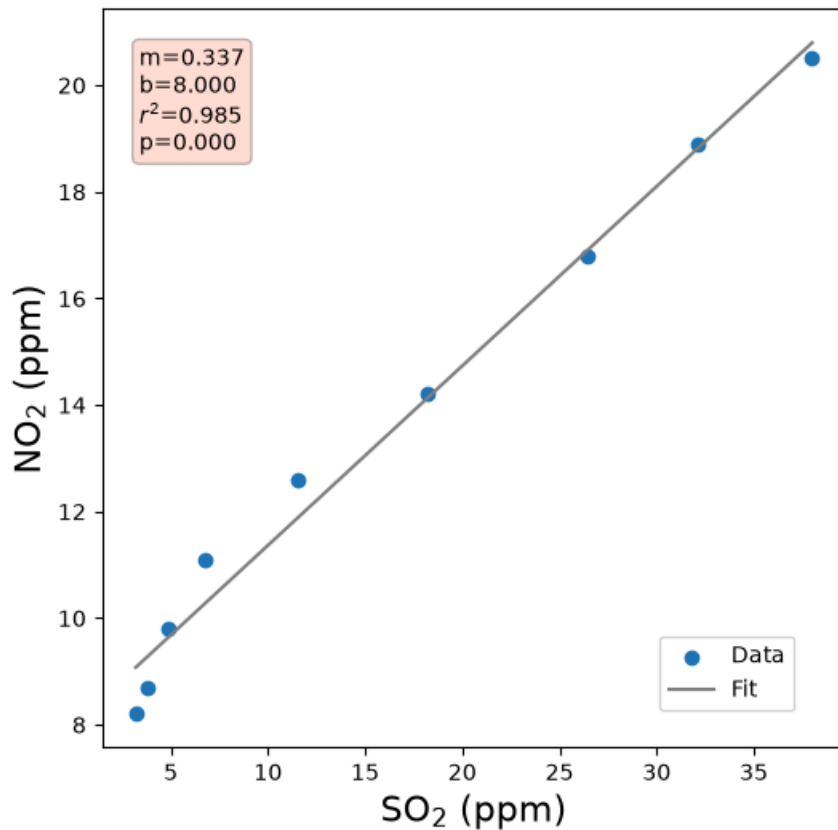
HW Introduction

The goal of today's homework is to get more practice line fitting and to better understand what the fitting parameters mean. Follow along the steps. Plus don't forget the last question.

Q1

- Make an excel file with at least 10 data points with x and y data.
- you can use data from another class, from the web etc.
- Aim for 10-50 data points
- DO NOT USE DATES. We will use dates later
- Read in the data plot the data and a best fit line.
- State where your data is from.
- My data is EPA summary data of SO2 and NO2 in air.
- Hint: In excel to keep things simple for now I would name your columns "x" and "y". Make sure all extra spaces are removed
- State if your correlation is signifant at the $p < 0.05$, $p < 0.01$, or $p < 0.001$ level in a figure caption.

```
Out[15]: <matplotlib.legend.Legend at 0x110899bd0>
```

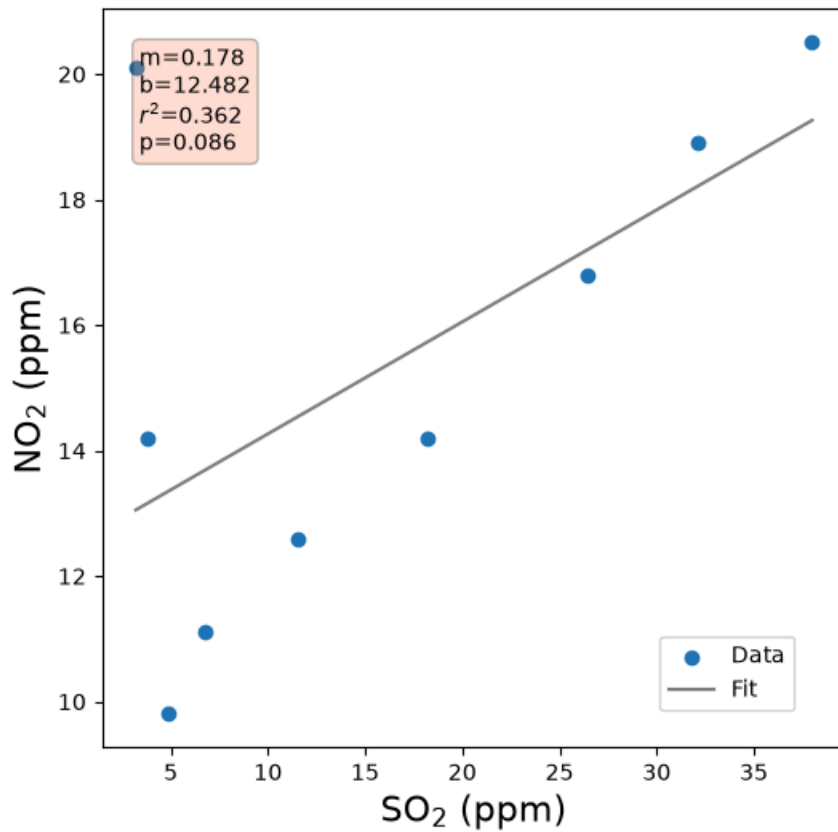


Q1 Figure Caption: SO₂ versus NO₂ concentrations from averaged annual values. The correlation is significant at the $p<0.001$ level. I used Gemini thinking 3.6 to help find the EPA data but I used no AI in the coding.

Q2

- Make a new sheet in your excel file that is a copy of your data.
- If you data had a significant correlation ($p<0.05$) change the data in the second sheet to make it not significant ($p>0.05$). If your data was insignificant change some values to make it significant.
- You need to manually adjust the data to change the correlation
- Now read in the second sheet using sheet_name keyword. Show the new correlation.
- plot your data and best fit line and state if your correlation is signifanct at the $p<0.05$, $p<0.01$, or $p<0.001$ level in a figure caption.
- Hint: I would name your new Sheet "Sheet2" with no spaces
- Hint: Save your excel file before reading in so changes show up

```
Out[16]: <matplotlib.legend.Legend at 0x110922050>
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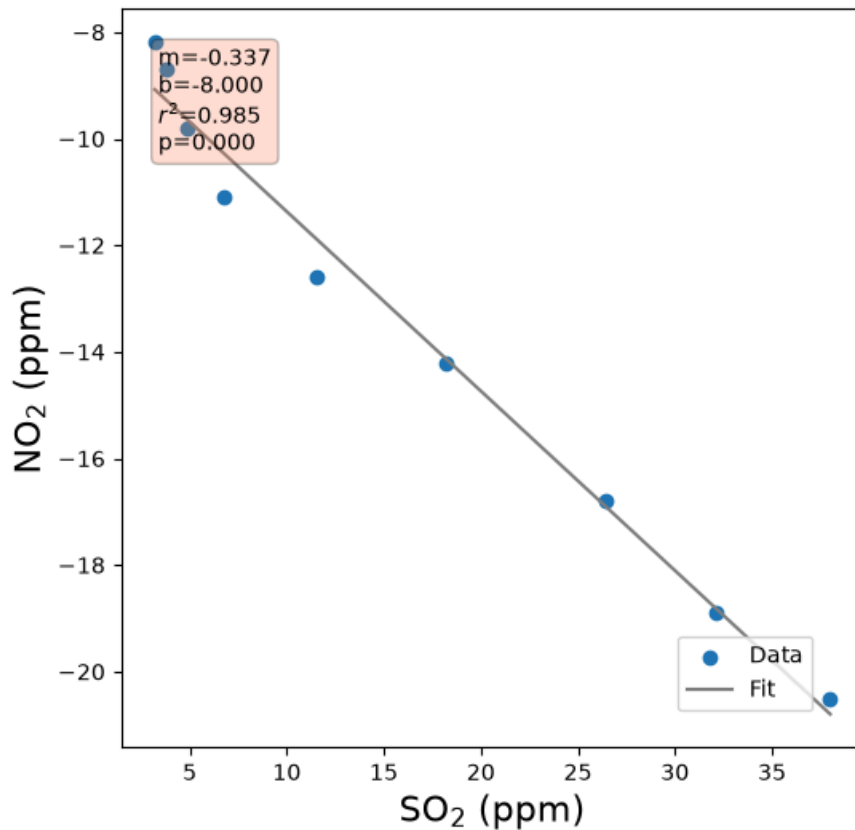


Q2 Figure Caption: SO₂ versus NO₂ concentrations that were not altered. The correlation is not significant at the $p < 0.05$ level. No AI was used in this quesiton.

Question 3.

- Make a third sheet in your excel file.
- Change your data again but this time take your significant correlation and multiple the y-values by -1.
- Make sure to take your significant correlation.
- Read it in and plot the data and the best fit line
- In the figure caption compare the m,b,r,p values to the original graph. Hoes does the r-value, p-value, and r-squared value compare between the data and the data when you multiplied by -1. What does multiplying by -1 do? Be specific.

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Out[17]: <matplotlib.legend.Legend at 0x110429cd0>
```



Q3 Figure Caption:.....

final 10%.

- As part of the class you have to comment on three of the lectures by using the comments section of the blog over the course of the semester.
- You do NOT have to comment now.
- You will comment as the semester proceeds.
- Comments can be wide ranging but need to be designed to improve the teaching or the homework given how you learn. Would you change the notebook? Would you change the homework. Is there a hint I should have given. How can you help me improve your learning and future students learning. Previous comments have already helped me to streamline the packets and homeworks.
- But now to make it fun you have to write an ipython script that randomly chooses your three classes. You need to randomly choose 2 classes between classes 2 and 16 and 1 class between classes 17 and 23.
- There are at least two ways to do this; you need to use numpy
- State how you figured this out.

I need to comment on classes 13, 15, and 17

REMEMBER

- You need to turn in both your notebook and your excel file to get credit

